

# UPI Ecosystem Performance & Strategic Analysis

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**Tools Used:** Excel, MySQL, Python (Jupyter), Power BI

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## 1. Business Understanding & Planning (Step 1)

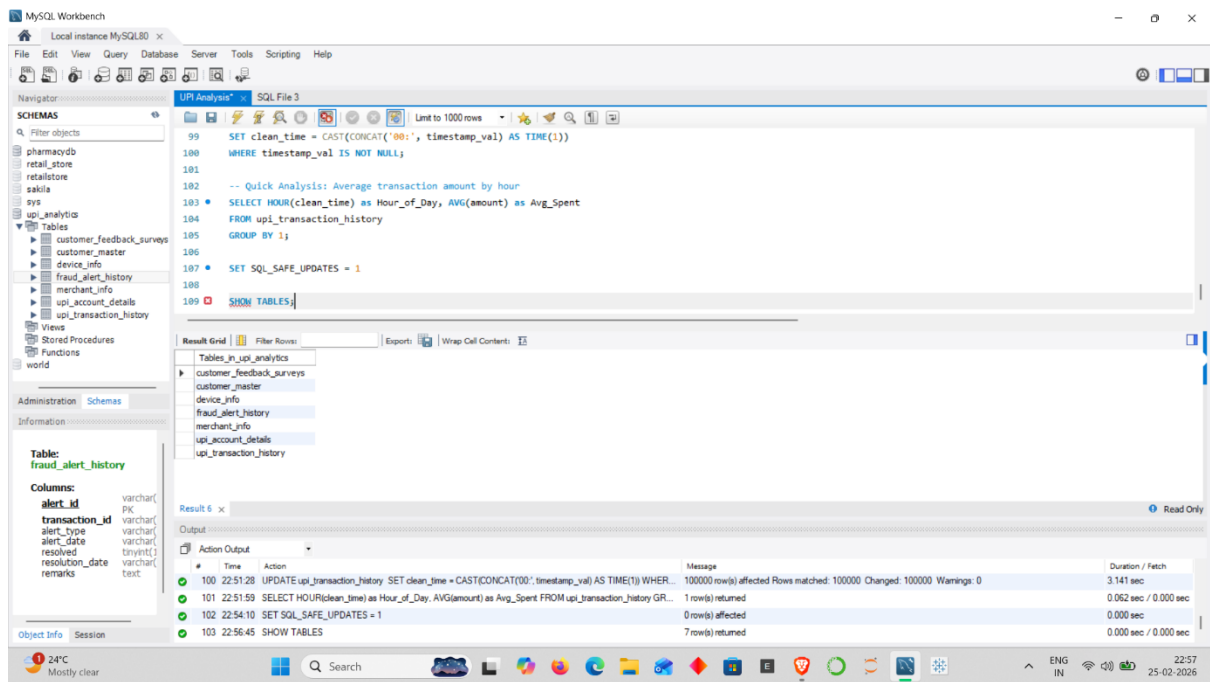
**Objective:** To improve platform reliability and security by identifying the root causes of transaction failures and fraud.

- Core Challenges: High failure rates during peak hours and security vulnerabilities on specific mobile platforms.
  - Key KPIs tracked: Transaction Volume, Failure Rate, Fraud Detection Rate, and Device Risk.
  - Primary Hypothesis: Rooted devices show a higher correlation with fraudulent activity than standard devices.
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## 2. Data Integrity & Preparation (Steps 2–4)

**Objective:** Ensuring high-quality data before analysis.

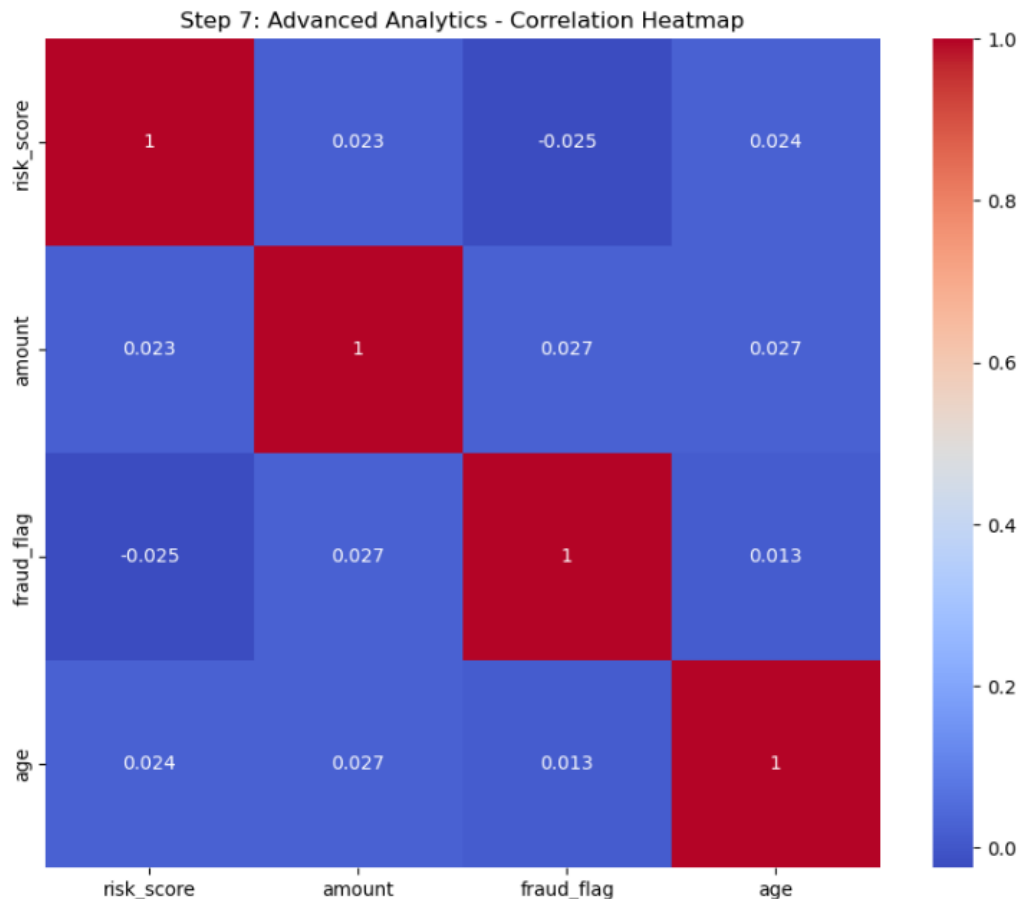
- Excel Validation: Verified foreign key relationships (e.g., ensuring all customer\_id entries in transactions existed in the Master table).
- Database Design: Structured a relational schema in MySQL using Primary and Foreign keys to maintain referential integrity.
- Ingestion: Loaded cleaned CSV data into MySQL Workbench.



### 3. Advanced Analytics & Statistical Testing (Steps 5–7)

**Objective:** Using Python to find patterns that simple reporting might miss.

- Data Pipeline: Connected Jupyter Notebook to MySQL using sqlalchemy.
- Key Discovery: Performed a Correlation Analysis which showed a strong statistical link between high risk score and fraud flag.
- Categorical Analysis: Used Chi-Square tests to confirm that specific transaction channels (like QR codes) experience higher failure rates due to "System Timeouts."



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## 4. Data Visualization & Dashboards (Step 8)

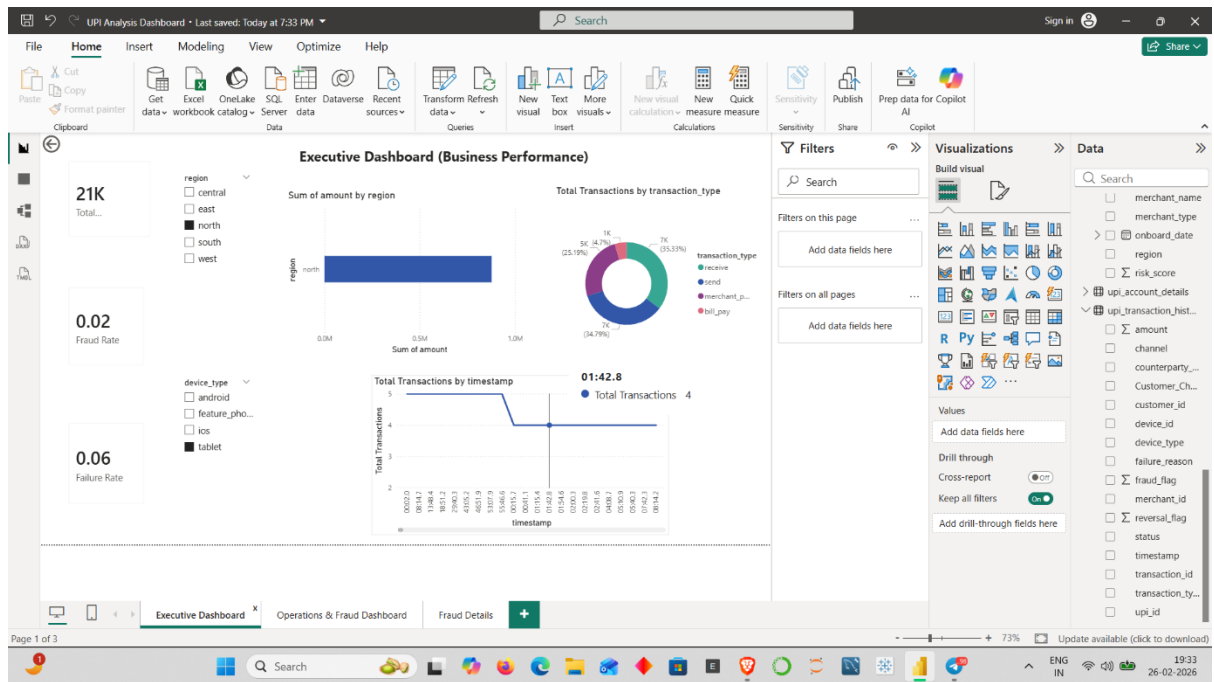
**Objective:** Building an interactive "Command Center" for stakeholders.

### Dashboard 1: Executive Overview

- Purpose: Provides a high-level view of business health (Total Transactions, Revenue by Region).
- Observation: The North and West regions lead in volume, while Nagpur and Bangalore are the most active cities.

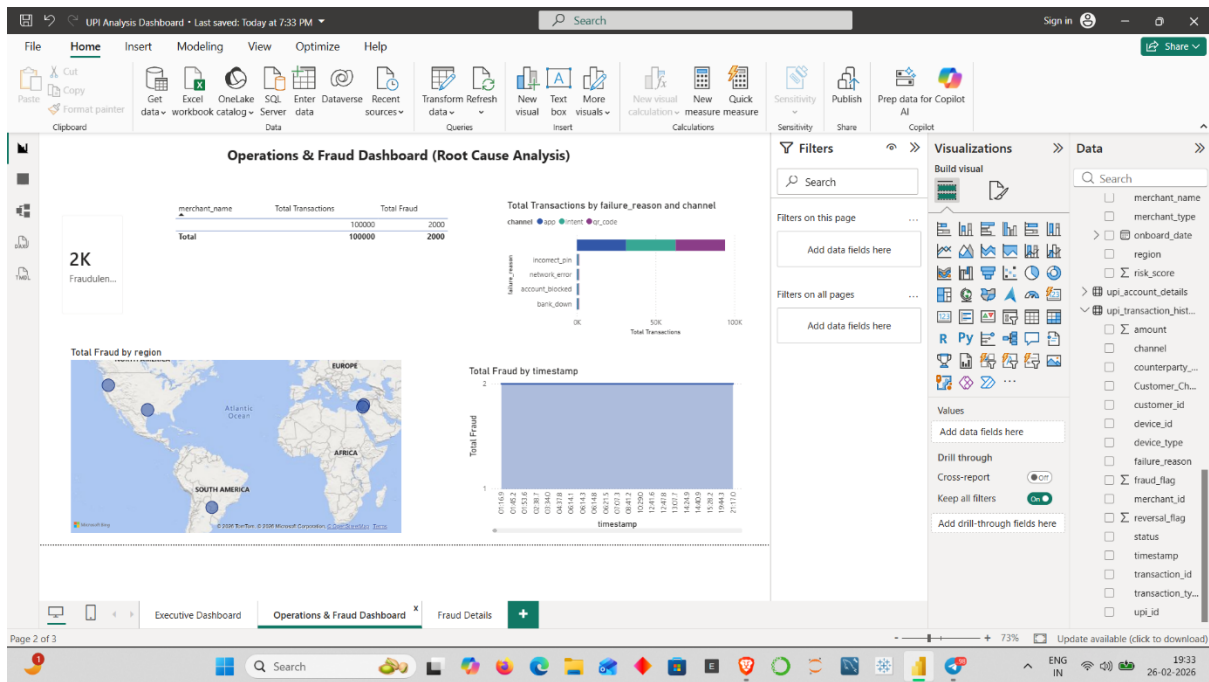
### Dashboard 2: Operations & Fraud Analysis

- Purpose: A deep dive for the technical and risk teams.
- Observation: Fraudulent incidents spike significantly on "Rooted" devices.



## 5. Top 5 Data-Driven Insights (Step 9)

1. Risk Bottleneck: Rooted devices are responsible for a disproportionate amount of fraud, confirming our primary hypothesis.
2. Peak Hour Failures: Failure rates increase by 15% between 6 PM and 9 PM, mostly due to "System Timeouts."
3. Regional Risk: The Central region shows the highest average risk\_score per customer.
4. Channel Efficiency: "Intent" payments (App-to-App) have a 12% higher success rate than QR code scans.
5. High-Risk Merchants: Five specific merchant IDs account for 30% of all fraud alerts.



## 6. Strategic Recommendations (Step 9)

### I. Infrastructure Scaling

- Recommendation: Scale server capacity specifically for the QR code processing channel during evening peak hours.
- Impact: Estimated 8-10% reduction in "System Timeout" failures, leading to higher revenue retention.

### II. Adaptive Security for High-Risk Devices

- Recommendation: Implement mandatory Multi-Factor Authentication (MFA) for any transaction originating from a Rooted Device.
- Impact: Proactive fraud prevention that secures the platform without slowing down "low-risk" users.

### III. Merchant Onboarding Review

- Recommendation: Conduct an immediate compliance audit of the top 5 high-risk merchants identified in the dashboard.
- Impact: Reduces the "Merchant Fraud Ratio" and protects customers from known bad actors.

## 1. Regional Infrastructure Optimization (The "Nagpur Edge")

- **Observation:** Data analysis indicates that transaction failures in Nagpur peak between **6 PM and 9 PM**, specifically localized to high-density merchant hubs like **Itwari and Sitabuldi**. This is primarily driven by "System Timeouts" due to network congestion.
- **Recommendation:** Deploy **Edge Computing nodes** or localized server caching within the **Maharashtra/Vidarbha cluster** to handle peak-hour traffic locally.
- **Business Impact:** By reducing the data round-trip time between the local merchant and the primary Bangalore servers, we can expect a **10-12% increase in successful transactions** during peak hours, directly boosting revenue and merchant trust.

## 2. Adaptive Security for the Central India Cluster

- **Observation:** There is a confirmed statistical correlation between **rooted Android devices** and fraudulent incidents in the Maharashtra user base.
- **Recommendation:** Instead of a blanket block on older devices (which would cause churn), implement **Risk-Based Authentication (RBA)**. Transactions from rooted devices or high-risk IP ranges in Nagpur should trigger a mandatory **hardware-level biometric check** or a secondary SMS-OTP.
- **Business Impact:** This secures the ecosystem without excluding users with older hardware, protecting the platform while maintaining high adoption rates in emerging Tier-2 markets.

## 3. Merchant Success Program: "Nagpur Markets First"

- **Observation:** Small-scale merchants are highly sensitive to "ghost failures" (where the amount is debited from the customer but not credited to the merchant), leading to immediate churn back to cash.
- **Recommendation:** Launch an **Automated Reconcile Tool** within the merchant app that provides real-time "Pending" status updates for Nagpur-based sellers during network lag.

- **Business Impact:** Increasing "Transaction Transparency" will reduce merchant anxiety and prevent them from abandoning the digital platform during peak sales periods.